



Echoes of HOMFLYPT in Kauffman

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1 Two Invariants

(HOMFLYPT 1986) (Oriented) link invariant $\mathbf{P}_L$:

$$\mathbf{P}_\circ = \frac{a - a^{-1}}{v - v^{-1}},$$

$$a\mathbf{P}_{\nearrow} - a^{-1}\mathbf{P}_{\searrow} = (v - v^{-1})\mathbf{P}_{\rangle\langle}.$$

(Jones–Ocneanu 1987)

Construction via traces on *Hecke algebras*

$$\mathbf{H}_n = \frac{\mathbf{Z}[v^{\pm 1}][Br_n]}{\langle \nearrow - \searrow - (v - v^{-1})\rangle \rangle \langle \rangle}.$$

A trace is a linear function that takes the same values on $\alpha\beta$ and $\beta\alpha$, for any elements α and β .

Br_n is the braid group

$$\left\langle \sigma_1, \dots, \sigma_{n-1} \left| \begin{array}{l} \text{Diagram 1} = \text{Diagram 2}, \\ \text{Diagram 3} = \text{Diagram 4} \end{array} \right. \right\rangle.$$

Unique family of traces $\text{tr}_n: H_n \rightarrow \mathbf{Q}(a, v)$ such that:

$$\begin{aligned} \text{tr}_1(1) &= \mathbf{P}_\circ, \\ \text{tr}_n(\beta) &= \mathbf{P}_\circ \text{tr}_{n-1}(\beta) \quad \text{for all } \beta \in H_{n-1}, \\ \text{tr}_n(\beta \sigma_{n-1}^{\pm 1}) &= -a^{\mp 1} \text{tr}_{n-1}(\beta) \quad \text{for all } \beta \in H_{n-1}. \end{aligned}$$

Now called *Markov traces*.

$$\mathbf{P}_{\hat{\beta}} = (-a)^e \text{tr}_n(\beta)$$

for $\beta \in Br_n$ of writhe e .

(Kauffman 1987) Independent(!) invariant $\mathbf{F}_L$.

$$\mathbf{F}_\circ = 1 + \frac{a - a^{-1}}{v - v^{-1}},$$

etc.

(Birman–Wenzl 1989, Murakami 1987)

Construction via traces on new algebras

$$\mathbf{C}_n = \mathbf{Z}[a^{\pm 1}, v^{\pm 1}][Br_n, e_1, \dots, e_{n-1}]/I_n,$$

where I_n is generated by

$$\begin{aligned} e_i^2 &= \mathbf{F}_\circ e_i, & e_i e_{i\pm 1} e_i &= e_i, \\ \sigma_i \sigma_{i\pm 1} e_i &= e_{i\pm 1} e_i, & \sigma_i e_i &= e_i \sigma_i = -a e_i, \\ e_i \sigma_{i\pm 1} e_i &= -a^{-1} e_i, & \sigma_i - \sigma_i^{-1} &= (v - v^{-1})(1 - e_i). \end{aligned}$$

Now called the *BMW algebras*.

$\mathbf{P}|_{a \rightarrow -v-N}$ is the $U_v(\mathfrak{sl}_N)$ link invariant;

$\mathbf{F}|_{a \rightarrow -v1-N}$ is the $U_v(\mathfrak{so}_N)$ link invariant.

Both specialize to Jones's polynomial: $\mathfrak{sl}_2 = \mathfrak{so}_3$.

For *torus knots* T :

- (Labastida–Pérez 1995)

$$\mathbf{F}_T^{\text{odd-}v} = \mathbf{P}_T.$$

- (T. 2026, inspired by Shelly 2016)

$$[a^{\text{low}}]\mathbf{F}_T^{\text{even-}v} = [a^{\text{low}}]\bar{\mathbf{P}}_T - [a^{\text{low}+2}]\bar{\mathbf{P}}_T,$$

where $\bar{\mathbf{P}}_T = \mathbf{P}_T/\mathbf{P}_\circ$.

Question How to find such “echoes” of $\mathbf{P}$ in $\mathbf{F}$?

Can brute-force via explicit formulas:

- $\mathbf{P}_T$ by (Jones 1987)
- $\mathbf{F}_T$ by (Y. Yokota 1991)

Example For the positive trefoil $T_{2,3}$,

$$\bar{\mathbf{P}}_{T_{2,3}} = a^2(v^2 + v^{-2}) - a^4,$$

$$\mathbf{F}_{T_{2,3}}^{\text{odd-}v} = \frac{(a^3 - a)(v^2 + v^{-2} - a^2)}{v - v^{-1}},$$

$$\begin{aligned} \mathbf{F}_{T_{2,3}}^{\text{even-}v} &= (a^2 - a^4)(v^2 + 1 + v^{-2}) + a^6 \\ &= a^2(v^2 + v^{-2} + 1) + a^4(\cdots). \end{aligned}$$

Questions But **why** such identities? And do they generalize beyond torus knots?

2 Two Traces

Pull back the HOMFLYPT trace along $C_n \twoheadrightarrow H_n$.

Lemma For $\delta \in \{0, 1\}$:

Unique family of traces $\text{tr}_n^\delta: C_n \rightarrow \mathbf{Q}(a, v)$ such that:

$$\begin{aligned} \text{tr}_1^\delta(1) &= \delta + \mathbf{P}_\circ, \\ \text{tr}_n^\delta(\beta) &= (\delta + \mathbf{P}_\circ) \text{tr}_{n-1}(\beta) \quad \text{for all } \beta \in C_{n-1}, \\ \text{tr}_n^\delta(\beta \sigma_{n-1}^{\pm 1}) &= -a^{\mp 1} \text{tr}_{n-1}(\beta) \quad \text{for all } \beta \in C_{n-1}, \\ \text{tr}_n^\delta(\beta e_{n-1}) &= \delta \text{tr}_{n-1}(\beta) \quad \text{for all } \beta \in C_{n-1}. \end{aligned}$$

$$(-a)^e \text{tr}_n^\delta(\beta) = \begin{cases} \mathbf{P}_{\hat{\beta}} & \delta = 0, \\ \mathbf{F}_{\hat{\beta}} & \delta = 1 \end{cases}$$

for $\beta \in Br_n$ of writhe e .

The torus knot $T_{m,n}$ is the closure of $\mathbf{c}_n^m$, where

$$\mathbf{c}_n = \sigma_1 \cdots \sigma_{n-1} \in Br_n.$$

Rewrite earlier identities using $\mathbf{c}_n^m$, tr_n^0 , tr_n^1 :

(Labastida–Pérez 1995)

$$\text{tr}_n^1(\mathbf{c}_n^m)^{\text{odd-}v} = \text{tr}_n^0(\mathbf{c}_n^m).$$

(T. 2026)

$$\begin{aligned} [a^{-n+1}] \text{tr}_n^1(\mathbf{c}_n^m) &= [a^{-n+1}] (\text{tr}_n^0(\mathbf{c}_n^m) + \overline{\text{tr}}_n^0(\mathbf{c}_n^m)) \\ &\quad - [a^{-n+3}] \overline{\text{tr}}_n^0(\mathbf{c}_n^m), \end{aligned}$$

where $\overline{\text{tr}}_n^0 = \text{tr}_n^0 / \mathbf{P}_\circ$.

We study these traces through character theory.

$\text{Irr}(C_n)$ is indexed by the partitions of the integers

$$n, \quad n-2, \quad n-4, \quad \dots$$

Get a Fourier decomposition:

$$\text{tr}_n^\delta(\beta) = \sum_{\substack{k \geq 0 \\ \lambda \vdash n-2k}} \omega_{n,\lambda}^\delta \cdot \chi^{n,\lambda}(\beta),$$

where $\omega_{n,\lambda}^\delta$ is called the *weight* of $\chi^{n,\lambda}$ in tr_n^δ .

Similarly, define $\overline{\omega}_{n,\lambda}^0$ to be its weight in $\overline{\text{tr}}_n^0$.

If $\lambda \vdash n$, then $\chi^{n,\lambda}$ factors through $C_n \twoheadrightarrow H_n$ and its values do not involve a .

(T. 2026) If $\lambda \vdash n$, then

$$\begin{aligned} [a^{-n+1}] \omega_{n,\lambda}^1 &= [a^{-n+1}] (\omega_{n,\lambda}^0 + \overline{\omega}_{n,\lambda}^0) \\ &\quad - [a^{-n+3}] \overline{\omega}_{n,\lambda}^0. \end{aligned}$$

Doesn't generalize to $|\lambda| < n$, where $\omega_{n,\lambda}^0, \overline{\omega}_{n,\lambda}^0 = 0$.
Nonetheless:

Definition Say that $\beta \in C_n$ is *clean* iff

$$[a^{-n+1}] (\omega_{n,\lambda}^1 \chi^{n,\lambda}(\beta)) = 0 \quad \text{for } |\lambda| < n.$$

Corollary (T. 2026) If $\beta \in Br_n$ is clean, then

$$\begin{aligned} [a^{-n+1}] \text{tr}_n^1(\beta) &= [a^{-n+1}] (\text{tr}_n^0(\beta) + \overline{\text{tr}}_n^0(\beta)) \\ &\quad - [a^{-n+3}] \overline{\text{tr}}_n^0(\beta). \end{aligned}$$

Expresses $[a^{\text{low}}] \mathbf{F}_{\hat{\beta}}$ in terms of $\mathbf{P}_{\hat{\beta}}$.

3 Cleanliness

(T. 2026) $\mathbf{c}_n^m$ is clean for *any* n and m .

That is, $\widehat{\mathbf{c}_n^m}$ can be any torus *link*.

Proof Key is that for $\lambda \vdash n - 2k$,

$$(\heartsuit) \quad \chi^{n,\lambda}(\mathbf{c}_n^m) = ((-a^2)^k v^{2\kappa(\lambda)})^{m/n} \chi^{n,\lambda}(\mathbf{c}_n^m)|_{v \rightarrow 1}$$

where κ is partition content and $(-)|_{v \rightarrow 1}$ is specialization to the Brauer algebra.

Remark Similar to Jones's trick to get $\text{tr}_n^0(\mathbf{c}_n^m)$.

Question Are there other clean braids?

Example $\sigma_1 \sigma_2^{-1} \in Br_3$ is not clean. But

$$\beta_\ell = \sigma_1 \sigma_2^{-1} (\mathbf{c}_n^n)^\ell$$

is clean for all $\ell > 0$!

By the way, $\mathbf{c}_n^n$ is the *full twist* on n strands.

$\widehat{\beta}_0$ is the unknot,

$\widehat{\beta}_1$ is the $(2, 5)$ torus knot,

$\widehat{\beta}_2$ is the $(-2, 3, 7)$ pretzel knot ...

Based on $\widehat{\beta}_0, \widehat{\beta}_1$, GPT-5.6 Sol realized that $(\heartsuit)$ gives:

(T. 2026) For any $n > 1$ and $\beta \in Br_n$:

$$\beta(\mathbf{c}_n^n)^\ell \text{ is clean for all } \ell \gg 0.$$

4 Future Work

Conjecture (Petrou–Hikami 2024)

For a general knot K , we have $\mathbf{F}_K^{\text{odd-}v} = \mathbf{P}_K$ if and only if the *Harer–Zagier transform*

$$\sum_j \frac{[a^j] \bar{\mathbf{P}}_K}{1 - \lambda v^j}$$

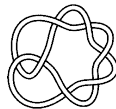
factors as a rational product of $(1 \pm \lambda q^k)$'s.

Question Which links L satisfy

$$[a^{\text{low}}] \mathbf{F}_L = [a^{\text{low}}](\mathbf{P}_L + \bar{\mathbf{P}}_L) - [a^{\text{low}+2}] \bar{\mathbf{P}}_L?$$

Sufficient that $L = \hat{\beta}$ for some clean braid β .

Fails for all non-torus knots of ≤ 8 crossings except...



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Thank you for listening.